Chapter 8

Numerical Solution of Ordinary Differential Equations

8.1 INTRODUCTION

Many problems in science and engineering can be reduced to the problem of solving differential equations satisfying certain given conditions. The analytical methods of solution, with which the reader is assumed to be familiar, can be applied to solve only a selected class of differential equations. Those equations which govern physical systems do not possess, in general closed-form solutions, and hence recourse must be made to numerical methods for solving such differential equations.

To describe various numerical methods for the solution of ordinary differential equations, we consider the general first order differential equation

$$\frac{dy}{dx} = f(x, y) \tag{8.1a}$$

with the initial condition,

$$y(x_0) = y_0 \tag{8.1b}$$

and illustrate the theory with respect to this equation. The methods so developed can, in general, be applied to the solution of systems of first-order equations, and will yield the solution in one of the two forms:

(i) A series for y in terms of powers of x, from which the value of y can be obtained by direct substitution.

(ii) A set of tabulated values of x and y.

The methods of Taylor and Picard belong to class (i), whereas those of Euler, Runge–Kutta, Adams–Bashforth, etc., belong to class (ii). These latter

methods are called *step-by-step* methods or *marching* methods because the values of y are computed by short steps ahead for equal intervals h of the independent variable. In the methods of Euler and Runge–Kutta, the interval length h should be kept small and hence these methods can be applied for tabulating y over a limited range only. If, however, the function values are desired over a wider range, the methods due to Adams–Bashforth, Adams–Moulton, Milne, etc., may be used. These methods use finite-differences and require 'starting values' which are usually obtained by Taylor's series or Runge–Kutta methods.

It is well-known that a differential equation of the nth order will have n arbitrary constants in its general solution. In order to compute the numerical solution of such an equation, we therefore need n conditions. Problems in which all the initial conditions are specified at the *initial* point only are called *initial value problems*. For example, the problem defined by Eqs. (8.1) is an *initial value problem*. On the other hand, in problems involving second-and higher-order differential equations, we may prescribe the conditions at two or more points. Such problems are called *boundary value problems*.

We shall first describe methods for solving initial value problems of the type (8.1), and at the end of the chapter we will outline methods for solving boundary value problems for second-order differential equations.

8.2 SOLUTION BY TAYLOR'S SERIES

We consider the differential equation

$$y' = f(x, y) \tag{8.1a}$$

with the initial condition

$$y(x_0) = y_0. \tag{8.1b}$$

If $y(x)$ is the exact solution of Eq. (8.1), then the Taylor's series for $y(x)$ around $x = x_0$ is given by

$$y(x) = y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2!} y_0'' + \cdots \tag{8.2}$$

If the values of $y_0', y_0'', \ldots$ are known, then Eq. (8.2) gives a power series for y. Using the formula for total derivatives, we can write

$$y'' = f' = f_x + y'f_y = f_x + ff_y,$$

where the suffixes denote partial derivatives with respect to the variable concerned. Similarly, we obtain

$$\begin{aligned} y''' = f'' &= f_{xx} + f_{xy}f + f(f_{yx} + f_{yy}f) + f_y(f_x + f_y f) \\ &= f_{xx} + 2ff_{xy} + f^2 f_{yy} + f_x f_y + ff_y^2 \end{aligned}$$

and other higher derivatives of y. The method can easily be extended to simultaneous and higher-order differential equations.

Example 8.1 From the Taylor series for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies

$$y' = x - y^2 \quad \text{and} \quad y(0) = 1.$$

The Taylor series for $y(x)$ is given by

$$y(x) = 1 + xy_0' + \frac{x^2}{2}y_0'' + \frac{x^3}{6}y_0''' + \frac{x^4}{24}y_0^{\text{iv}} + \frac{x^5}{120}y_0^{\text{v}} + \cdots$$

The derivatives $y_0', y_0'', \ldots$ etc. are obtained thus:

$$y'(x) = x - y^2 \qquad y_0' = -1$$

$$y''(x) = 1 - 2yy' \qquad y_0'' = 3$$

$$y'''(x) = -2yy'' - 2y'^2 \qquad y_0''' = -8$$

$$y^{\text{iv}}(x) = -2yy''' - 6y'y'' \qquad y_0^{\text{iv}} = 34$$

$$y^{\text{v}}(x) = -2yy^{\text{iv}} - 8y'y''' - 6y''^2 \qquad y_0^{\text{v}} = -186$$

Using these values, the Taylor series becomes

$$y(x) = 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 - \frac{31}{20}x^5 + \cdots$$

To obtain the value of $y(0.1)$ correct to four decimal places, it is found that the terms up to x^4 should be considered, and we have $y(0.1) = 0.9138.$

Suppose that we wish to find the range of values of x for which the above series, truncated after the term containing x^4, can be used to compute the values of y correct to four decimal places. We need only to write

$$\frac{31}{20}x^5 \le 0.00005 \quad \text{or} \quad x \le 0.126.$$

Example 8.2 Given the differential equation

$$y'' - xy' - y = 0$$

with the conditions $y(0) = 1$ and $y'(0) = 0$, use Taylor's series method to determine the value of $y(0.1)$.

We have $y(x) = 1$ and $y'(x) = 0$ when $x = 0$. The given differential equation is

$$y''(x) = xy'(x) + y(x) \tag{i}$$

Hence $y''(0) = y(0) = 1$. Successive differentiation of (i) gives

$$y'''(x) = xy''(x) + y'(x) + y'(x) = xy''(x) + 2y'(x), \tag{ii}$$

$$y^{\text{iv}}(x) = xy'''(x) + y''(x) + 2y''(x) = xy'''(x) + 3y''(x), \tag{iii}$$

$$y^{\text{v}}(x) = xy^{\text{iv}}(x) + y'''(x) + 3y'''(x) = xy^{\text{iv}}(x) + 4y'''(x), \tag{iv}$$

$$y^{\text{vi}}(x) = xy^{\text{v}}(x) + y^{\text{iv}}(x) + 4y^{\text{iv}}(x) = xy^{\text{v}}(x) + 5y^{\text{iv}}(x), \tag{v}$$

and similarly for higher derivatives. Putting $x = 0$ in (ii) to (v), we obtain

$$y'''(0) = 2y'(0) = 0, \quad y^{iv}(0) = 3y''(0) = 3, \quad y^{v}(0) = 0, \quad y^{vi}(0) = 5.$$

By Taylor's series, we have

$$y(x) = y(0) + xy'(0) + \frac{x^2}{2}y''(0) + \frac{x^3}{6}y'''(0) + \frac{x^4}{24}y^{iv}(0) + \frac{x^5}{120}y^{v}(0) + \frac{x^6}{720}y^{vi}(0) + \cdots$$

Hence

$$y(0.1) = 1 + \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24}(3) + \frac{(0.1)^6}{720}(5) + \cdots$$

$$= 1 + 0.005 + 0.0000125, \text{ neglecting the last term}$$

$$= 1.0050125, \text{ correct to seven decimal places.}$$

8.3 PICARD'S METHOD OF SUCCESSIVE APPROXIMATIONS

Integrating the differential equation given in Eq. (8.1), we obtain

$$y = y_0 + \int_{x_0}^{x} f(x, y)\, dx. \tag{8.3}$$

Equation (8.3), in which the unknown function y appears under the integral sign, is called an *integral equation*. Such an equation can be solved by the method of successive approximations in which the first approximation to y is obtained by putting y_0 for y on right side of Eq. (8.3), and we write

$$y^{(1)} = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx$$

The integral on the right can now be solved and the resulting $y^{(1)}$ is substituted for y in the integrand of Eq. (8.3) to obtain the second approximation $y^{(2)}$:

$$y^{(2)} = y_0 + \int_{x_0}^{x} f(x, y^{(1)})\, dx$$

Proceeding in this way, we obtain $y^{(3)}, y^{(4)}, \ldots, y^{(n-1)}$ and $y^{(n)}$, where

$$y^{(n)} = y_0 + \int_{x_0}^{x} f(x, y^{(n-1)})\, dx \qquad \text{with } y^{(0)} = y_0 \tag{8.4}$$

Hence this method yields a sequence of approximations $y^{(1)}, y^{(2)}, \ldots, y^{(n)}$ and it can be proved (*see*, for example, the book by Levy and Baggot) that if the function $f(x, y)$ is bounded in some region about the point (x_0, y_0) and if $f(x, y)$ satisfies the *Lipschitz condition*, viz.,

$$|f(x, y) - f(x, \bar{y})| \le K |y - \bar{y}| \qquad K \text{ being a constant} \tag{8.5}$$

then the sequence $y^{(1)}, y^{(2)}, \ldots$ converges to the solution of Eq. (8.1).

Example 8.3 Solve the equation $y' = x + y^2$, subject to the condition $y = 1$ when $x = 0$.

We start with $y^{(0)} = 1$ and obtain

$$y^{(1)} = 1 + \int_0^x (x+1)\,dx = 1 + x + \frac{1}{2}x^2.$$

Then the second approximation is

$$y^{(2)} = 1 + \int_0^x \left[x + \left(1 + x + \frac{1}{2}x^2 \right)^2 \right] dx$$

$$= 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5.$$

It is obvious that the integrations might become more and more difficult as we proceed to higher approximations.

Example 8.4 Given the differential equation

$$\frac{dy}{dx} = \frac{x^2}{y^2 + 1}$$

with the initial condition $y = 0$ when $x = 0$, use Picard's method to obtain y for $x = 0.25$, 0.5 and 1.0 correct to three decimal places.

We have

$$y = \int_0^x \frac{x^2}{y^2 + 1}\,dx.$$

Setting $y^{(0)} = 0$, we obtain

$$y^{(1)} = \int_0^x x^2\,dx = \frac{1}{3}x^3$$

and

$$y^{(2)} = \int_0^x \frac{x^2}{(1/9)\,x^6 + 1}\,dx = \tan^{-1}\left(\frac{1}{3}x^3 \right) = \frac{1}{3}x^3 - \frac{1}{81}x^9 + \cdots$$

so that $y^{(1)}$ and $y^{(2)}$ agree to the first term, viz., $(1/3)x^3$. To find the range of values of x so that the series with the term $(1/3)x^3$ alone will give the result correct to three decimal places, we put

$$\frac{1}{81}x^9 \le 0.0005$$

which yields

$$x \le 0.7$$

Hence

$$y(0.25) = \frac{1}{3}(0.25)^3 = 0.005$$

$$y(0.5) = \frac{1}{3}(0.5)^3 = 0.042$$

$$y(1.0) = \frac{1}{3} - \frac{1}{81} = 0.321$$

8.4 EULER'S METHOD

We have so far discussed the methods which yield the solution of a differential equation in the form of a power series. We will now describe the methods which give the solution in the form of a set of tabulated values.

Suppose that we wish to solve the Eqs. (8.1) for values of y at $x = x_r = x_0 + rh (r = 1, 2, \ldots)$. Integrating Eq. (8.1), we obtain

$$y_1 = y_0 + \int_{x_0}^{x_1} f(x, y)\,dx. \tag{8.6}$$

Assuming that $f(x, y) = f(x_0, y_0)$ in $x_0 \le x \le x_1$, this gives Euler's formula

$$y_1 \approx y_0 + hf(x_0, y_0). \tag{8.7a}$$

Similarly for the range $x_1 \le x \le x_2$, we have

$$y_2 = y_1 + \int_{x_1}^{x_2} f(x, y)\,dx.$$

Substituting $f(x_1, y_1)$ for $f(x, y)$ in $x_1 \le x \le x_2$ we obtain

$$y_2 \approx y_1 + hf(x_1, y_1). \tag{8.7b}$$

Proceeding in this way, we obtain the general formula

$$y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \ldots \tag{8.8}$$

The process is very slow and to obtain reasonable accuracy with Euler's method, we need to take a smaller value for h. Because of this restriction

on h, the method is unsuitable for practical use and a modification of it, known as the *modified Euler method*, which gives more accurate results, will be described in Section 8.4.2.

Example 8.5 To illustrate Euler's method, we consider the differential equation $y' = -y$ with the condition $y(0) = 1$.

Successive application of Eq. (8.8) with $h = 0.01$ gives

$$y(0.01) = 1 + 0.1(-1) = 0.99$$

$$y(0.02) = 0.99 + 0.01(-0.99) = 0.9801$$

$$y(0.03) = 0.9801 + 0.01(-0.9801) = 0.9703$$

$$y(0.04) = 0.9703 + 0.01(-0.9703) = 0.9606.$$

The exact solution is $y = e^{-x}$ and from this the value at $x = 0.04$ is 0.9608.

8.4.1 Error Estimates for the Euler Method

Let the true solution of the differential equation at $x = x_n$ be $y(x_n)$ and also let the approximate solution be y_n. Now, expanding $y(x_{n+1})$ by Taylor's series, we get

$$\begin{aligned} y(x_{n+1}) &= y(x_n) + hy'(x_n) + \frac{h^2}{2} y''(x_n) + \cdots \\ &= y(x_n) + hy'(x_n) + \frac{h^2}{2} y''(\tau_n), \qquad \text{where } x_n \le \tau_n \le x_{n+1}. \end{aligned} \tag{8.9}$$

We usually encounter two types of errors in the solution of differential equations. These are (i) local errors, and (ii) rounding errors. The local error is the result of replacing the given differential equation by means of the equation

$$y_{n+1} = y_n + hy'_n.$$

This error is given by

$$L_{n+1} = -\frac{1}{2} h^2 y''(\tau_n) \tag{8.10}$$

The total error is then defined by

$$e_n = y_n - y(x_n) \tag{8.11}$$

Since y_0 is exact, it follows that $e_0 = 0$.

Neglecting the rounding error, we write the total solution error as

$$\begin{aligned} e_{n+1} &= y_{n+1} - y(x_{n+1}) \\ &= y_n + hy'_n - [y(x_n) + hy'(x_n) - L_{n+1}] \\ &= e_n + h[f(x_n, y_n) - y'(x_n)] + L_{n+1}. \end{aligned}$$

$$\Rightarrow \qquad e_{n+1} = e_n + h\left[f(x_n, y_n) - f(x_n, y(x_n))\right] + L_{n+1}.$$

By mean value theorem, we write

$$f(x_n, y_n) - f(x_n, y(x_n)) = \left[y_n - y(x_n)\right]\frac{\partial f}{\partial y}(x_n, \xi_n), \quad y(x_n) \le \xi_n \le y_n.$$

Hence, we have

$$e_{n+1} = e_n\left[1 + hf_y(x_n, \xi_n)\right] + L_{n+1} \tag{8.12}$$

Since $e_0 = 0$, we obtain successively:

$$e_1 = L_1; \quad e_2 = \left[1 + hf_y(x_1, \xi_1)\right]L_1 + L_2;$$

$$e_3 = \left[1 + hf_y\,(x_2, \xi_2)\right]\left[1 + hf_y(x_1, \xi_1)\right](L_1 + L_2) + L_3; \text{ etc.}$$

See the book by Isaacson and Keller [1966] for more details.

Example 8.6 We consider, again, the differential equation $y' = -y$ with the condition $y(0) = 1,$ which we have solved by Euler's method in Example 8.5.

Choosing $h = 0.01,$ we have

$$1 + hf_y(x_n, \xi_n) = 1 + 0.01(-1) = 0.99.$$

and

$$L_{n+1} = -\frac{1}{2}h^2 y''(\rho_n) = -0.00005\, y(\rho_n).$$

In this problem, $y(\rho_n) \le y(x_n),$ since y' is negative. Hence we successively obtain

$$|L_1| \le 0.00005 = 5\times 10^{-5},$$

$$|L_2| \le (0.00005)\,(0.99) < 5\times 10^{-5},$$

$$|L_3| \le (0.00005)\,(0.9801) < 5\times 10^{-5},$$

and so on. For computing the total solution error, we need an estimate of the rounding error. If we neglect the rounding error, i.e., if we set

$$R_{n+1} = 0,$$

then using the above bounds, we obtain from Eq. (8.12) the estimates

$$e_0 = 0,$$

$$|e_1| \le 5\times 10^{-5}$$

$$|e_2| \le 0.99e_1 + 5\times 10^{-5} < 10^{-4}$$

$$|e_3| \le 0.99e_2 + 5\times 10^{-5} < 10^{-4} + 5\times 10^{-5}$$

$$|e_4| \le 0.99e_3 + 5\times 10^{-5} < 10^{-4} + 10^{-4} = 2\times 10^{-4} = 0.0002$$

$$\vdots$$

It can be verified that the estimate for e_4 agrees with the actual error in the value of $y(0.04)$ obtained in Example 8.5.

8.4.2 Modified Euler's Method

Instead of approximating $f(x, y)$ by $f(x_0, y_0)$ in Eq. (8.6), we now approximate the integral given in Eq. (8.6) by means of trapezoidal rule to obtain

$$y_1 = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)] \tag{8.13}$$

We thus obtain the iteration formula

$$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})], \quad n = 0, 1, 2, \ldots \tag{8.14}$$

where $y_1^{(n)}$ is the nth approximation to y_1. The iteration formula (8.14) can be started by choosing $y_1^{(0)}$ from Euler's formula:

$$y_1^{(0)} = y_0 + hf(x_0, y_0).$$

Example 8.7 Determine the value of y when $x = 0.1$ given that

$$y(0) = 1 \quad \text{and} \quad y' = x^2 + y$$

We take $h = 0.05$. With $x_0 = 0$ and $y_0 = 1.0$, we have $f(x_0, y_0) = 1.0$. Hence Euler's formula gives

$$y_1^{(0)} = 1 + 0.05(1) = 1.05$$

Further, $x_1 = 0.05$ and $f(x_1, y_1^{(0)}) = 1.0525$. The average of $f(x_0, y_0)$ and $f(x_1, y_1^{(0)})$ is 1.0262. The value of $y_1^{(1)}$ can therefore be computed by using Eq. (8.14) and we obtain

$$y_1^{(1)} = 1.0513.$$

Repeating the procedure, we obtain $y_1^{(2)} = 1.0513$. Hence we take $y_1 = 1.0513$, which is correct to four decimal places.

Next, with $x_1 = 0.05$, $y_1 = 1.0513$ and $h = 0.05$, we continue the procedure to obtain y_2, i.e., the value of y when $x = 0.1$. The results are

$$y_2^{(0)} = 1.1040, \quad y_2^{(1)} = 1.1055, \quad y_2^{(2)} = 1.1055.$$

Hence we conclude that the value of y when $x = 0.1$ is 1.1055.

8.5 RUNGE–KUTTA METHODS

As already mentioned, Euler's method is less efficient in practical problems since it requires h to be small for obtaining reasonable accuracy. The

Runge–Kutta methods are designed to give greater accuracy and they possess the advantage of requiring only the function values at some selected points on the subinterval.

If we substitute $y_1 = y_0 + hf(x_0, y_0)$ on the right side of Eq. (8.13), we obtain

$$y_1 = y_0 + \frac{h}{2}[f_0 + f(x_0 + h, y_0 + hf_0)],$$

where $f_0 = f(x_0, y_0)$. If we now set

$$k_1 = hf_0 \quad \text{and} \quad k_2 = hf(x_0 + h,\ y_0 + k_1)$$

then the above equation becomes

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2), \tag{8.15}$$

which is the *second-order Runge–Kutta* formula. The error in this formula can be shown to be of order h^3 by expanding both sides by Taylor's series. Thus, the left side gives

$$y_0 + hy_0' + \frac{h^2}{2}y_0'' + \frac{h^3}{6}y_0''' + \cdots$$

and on the right side

$$k_2 = hf\ (x_0 + h, y_0 + hf_0) = h\left[f_0 + h\frac{\partial f}{\partial x_0} + hf_0\frac{\partial f}{\partial y_0} + \mathrm{O}(h^2)\right].$$

Since

$$\frac{df\,(x, y)}{dx} = \frac{\partial f}{\partial x} + f\frac{\partial f}{\partial y},$$

we obtain

$$k_2 = h[f_0 + hf_0' + \mathrm{O}(h^2)] = hf_0 + h^2 f_0' + \mathrm{O}(h^3),$$

so that the right side of Eq. (8.15) gives

$$y_0 + \frac{1}{2}[hf_0 + hf_0 + h^2 f_0' + \mathrm{O}(h^3)] = y_0 + hf_0 + \frac{1}{2}h^2 f_0' + \mathrm{O}(h^3)$$

$$= y_0 + hy_0' + \frac{h^2}{2}y_0'' + \mathrm{O}(h^3).$$

It therefore follows that the Taylor series expansions of both sides of Eq. (8.15) agree up to terms of order h^2, which means that the error in this formula is of order h^3.

More generally, if we set

$$y_1 = y_0 + W_1k_1 + W_2k_2 \tag{8.16a}$$

where

$$\left.\begin{aligned} k_1 &= hf_0 \\ k_2 &= hf\ (x_0 + \alpha_0 h,\ y_0 + \beta_0 k_1) \end{aligned}\right\} \tag{8.16b}$$

then the Taylor series expansions of both sides of the last equation in (8.16a) gives the identity

$$y_0 + hf_0 + \frac{h^2}{2}\left(\frac{\partial f}{\partial x} + f_0\frac{\partial f}{\partial y}\right) + \mathrm{O}(h^3) = y_0 + (W_1 + W_2)hf_0 + W_2h^2\left(\alpha_0\frac{\partial f}{\partial x} + \beta_0 f_0\frac{\partial f}{\partial y}\right) + \mathrm{O}(h^3).$$

Equating the coefficients of $f(x, y)$ and its derivatives on both sides, we obtain the relations

$$W_1 + W_2 = 1, \quad W_2\alpha_0 = \frac{1}{2}, \quad W_2\beta_0 = \frac{1}{2}. \tag{8.17}$$

Clearly, $\alpha_0 = \beta_0$ and if α_0 is assigned any value arbitrarily, then the remaining parameters can be determined uniquely. If we set, for example, $\alpha_0 = \beta_0 = 1$, then we immediately obtain $W_1 = W_2 = 1/2$, which gives formula (8.15).

It follows, therefore, that there are several second-order Runge–Kutta formulae and that formulae (8.16) and (8.17) constitute just one of several such formulae.

Higher-order Runge–Kutta formulae exist, of which we mention only the *fourth-order formula* defined by

$$y_1 = y_0 + W_1k_1 + W_2k_2 + W_3k_3 + W_4k_4 \tag{8.18a}$$

where

$$\left.\begin{aligned} k_1 &= hf(x_0, y_0) \\ k_2 &= hf(x_0 + \alpha_0 h, y_0 + \beta_0 k_1) \\ k_3 &= hf(x_0 + \alpha_1 h, y_0 + \beta_1 k_1 + \nu_1 k_2) \\ k_4 &= hf(x_0 + \alpha_2 h, y_0 + \beta_2 k_1 + \nu_2 k_2 + \delta_1 k_3), \end{aligned}\right\} \tag{8.18b}$$

where the parameters have to be determined by expanding both sides of the first equation of (8.18a) by Taylor's series and securing agreement of terms up to and including those containing h^4. The choice of the parameters is, again, arbitrary and we have therefore several fourth-order Runge–Kutta formulae. If, for example, we set

$$\left.\begin{aligned} &\alpha_0 = \beta_0 = \frac{1}{2}, && \alpha_1 = \frac{1}{2}, && \alpha_2 = 1, \\ &\beta_1 = \frac{1}{2}(\sqrt{2} - 1), && \beta_2 = 0 \\ &\nu_1 = 1 - \frac{1}{\sqrt{2}}, && \nu_2 = -\frac{1}{\sqrt{2}}, && \delta_1 = 1 + \frac{1}{\sqrt{2}}, \\ &W_1 = W_4 = \frac{1}{6}, && W_2 = \frac{1}{3}\left(1 - \frac{1}{\sqrt{2}}\right), && W_3 = \frac{1}{3}\left(1 + \frac{1}{\sqrt{2}}\right), \end{aligned}\right\} \tag{8.19}$$

we obtain the method of Gill, whereas the choice

$$\left.\begin{aligned} &\alpha_0 = \alpha_1 = \frac{1}{2}, && \beta_0 = \nu_1 = \frac{1}{2} \\ &\beta_1 = \beta_2 = \nu_2 = 0, && \alpha_2 = \delta_1 = 1 \\ &W_1 = W_4 = \frac{1}{6}, && W_2 = W_3 = \frac{2}{6} \end{aligned}\right\} \tag{8.20}$$

leads to the fourth-order Runge–Kutta formula, the most commonly used one in practice:

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{8.21a}$$

where

$$\left.\begin{aligned} k_1 &= hf(x_0, y_0) \\ k_2 &= hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) \\ k_3 &= hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) \\ k_4 &= hf(x_0 + h, y_0 + k_3) \end{aligned}\right\} \tag{8.21b}$$

in which the error is of order h^5. Complete derivation of the formula is exceedingly complicated, and the interested reader is referred to the book by Levy and Baggot. We illustrate here the use of the fourth-order formula by means of examples.

Example 8.8 Given $dy/dx = y - x$ where $y(0) = 2$, find $y(0.1)$ and $y(0.2)$ correct to four decimal places.

(i) *Runge–Kutta second-order formula:* With $h = 0.1$, we find $k_1 = 0.2$ and $k_2 = 0.21$. Hence

$$y_1 = y(0.1) = 2 + \frac{1}{2}(0.41) = 2.2050.$$

To determine $y_2 = y(0.2)$, we note that $x_0 = 0.1$ and $y_0 = 2.2050$. Hence, $k_1 = 0.1(2.105) = 0.2105$ and $k_2 = 0.1(2.4155 - 0.2) = 0.22155$.

It follows that

$$y_2 = 2.2050 + \frac{1}{2}(0.2105 + 0.22155) = 2.4210.$$

Proceeding in a similar way, we obtain

$$y_3 = y(0.3) = 2.6492 \quad \text{and} \quad y_4 = y(0.4) = 2.8909$$

We next choose $h = 0.2$ and compute $y(0.2)$ and $y(0.4)$ directly. With $h = 0.2$. $x_0 = 0$ and $y_0 = 2$, we obtain $k_1 = 0.4$ and $k_2 = 0.44$ and hence $y(0.2) = 2.4200$. Similarly, we obtain $y(0.4) = 2.8880$.

From the analytical solution $y = x + 1 + e^x$, the exact values of $y(0.2)$ and $y(0.4)$ are respectively 2.4214 and 2.8918. To study the order of convergence of this method, we tabulate the values as follows:

x	*Computed y*	*Exact y*	*Difference*	*Ratio*
0.2	$h = 0.1$: 2.4210	2.4214	0.0004	3.5
	$h = 0.2$: 2.4200		0.0014	
0.4	$h = 0.1$: 2.8909	2.8918	0.0009	4.2
	$h = 0.2$: 2.8880		0.0038	

It follows that the method has an h^2-order of convergence.

(ii) *Runge–Kutta fourth-order formula:* To determine $y(0.1)$, we have $x_0 = 0$, $y_0 = 2$ and $h = 0.1$. We then obtain

$$k_1 = 0.2,$$
$$k_2 = 0.205$$
$$k_3 = 0.20525$$
$$k_4 = 0.21053.$$

Hence

$$y(0.1) = 2 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 2.2052.$$

Proceeding similarly, we obtain $y(0.2) = 2.4214$.

Example 8.9 Given $dy/dx = 1 + y^2$, where $y = 0$ when $x = 0$, find $y(0.2)$, $y(0.4)$ and $y(0.6)$.

We take $h = 0.2$. With $x_0 = y_0 = 0$, we obtain from (8.21a) and (8.21b),

$$k_1 = 0.2,$$
$$k_2 = 0.2\,(1.01) = 0.202,$$
$$k_3 = 0.2\,(1 + 0.010201) = 0.20204,$$
$$k_4 = 0.2\,(1 + 0.040820) = 0.20816,$$

and

$$y(0.2) = 0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.2027,$$

which is correct to four decimal places.

To compute $y(0.4)$, we take $x_0 = 0.2$, $y_0 = 0.2027$ and $h = 0.2$. With these values, Eqs. (8.21a) and (8.21b) give

$$k_1 = 0.2\,[1 + (0.2027)^2] = 0.2082,$$

$$k_2 = 0.2\,[1 + (0.3068)^2] = 0.2188,$$

$$k_3 = 0.2\,[1 + (0.3121)^2] = 0.2195,$$

$$k_4 = 0.2\,[1 + (0.4222)^2] = 0.2356,$$

and

$$y(0.4) = 0.2027 + 0.2201 = 0.4228,$$

correct to four decimal places.

Finally, taking $x_0 = 0.4$, $y_0 = 0.4228$ and $h = 0.2$, and proceeding as above, we obtain $y(0.6) = 0.6841$.

Example 8.10 We consider the initial value problem $y' = 3x + y/2$ with the condition $y(0) = 1$.

The following table gives the values of $y(0.2)$ by different methods, the exact value being 1.16722193. It is seen that the *fourth-order* Runge–Kutta method gives the accurate value for $h = 0.05$.

Method	*h*	*Computed value*
Euler	0.2	1.100 000 00
	0.1	1.132 500 00
	0.05	1.149 567 58
Modified Euler	0.2	1.100 000 00
	0.1	1.150 000 00
	0.05	1.162 862 42
Fourth-order Runge–Kutta	0.2	1.167 220 83
	0.1	1.167 221 86
	0.05	1.167 221 93

8.6 PREDICTOR–CORRECTOR METHODS

In the methods described so far, to solve a differential equation over a single interval, say from $x = x_n$ to $x = x_{n+1}$, we required information only at the beginning of the interval, i.e. at $x = x_n$. *Predictor–corrector* methods are the ones which require function values at x_n, x_{n-1}, x_{n-2}, … for the computation of the function value at x_{n+1} A *predictor* formula is used to predict the value of y at x_{n+1} and then a *corrector* formula is used to improve the value of y_{n+1}.

In Section 8.6.1 we derive Predictor–corrector formulae which use backward differences and in Section 8.6.2 we describe Milne's method which uses forward differences.

8.6.1 Adams–Moulton Method

Newton's backward difference interpolation formula can be written as

$$f(x, y) = f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \frac{n(n+1)(n+2)}{6}\nabla^3 f_0 + \cdots \qquad (8.22)$$

where

$$n = \frac{x - x_0}{h} \quad \text{and} \quad f_0 = f(x_0, y_0).$$

If this formula is substituted in

$$y_1 = y_0 + \int_{x_0}^{x_1} f(x, y)\, dx, \qquad (8.23)$$

we get

$$y_1 = y_0 + \int_{x_0}^{x_1} \left[f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \cdots \right] dx$$

$$= y_0 + h \int_0^1 \left[f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \cdots \right] dn$$

$$= y_0 + h\left(1 + \frac{1}{2}\nabla + \frac{5}{12}\nabla^2 + \frac{3}{8}\nabla^3 + \frac{251}{720}\nabla^4 + \cdots\right) f_0.$$

It can be seen that the right side of the above relation depends only on y_0, y_{-1}, y_{-2}, …, all of which are known. Hence this formula can be used to compute y_1. We therefore write it as

$$y_1^{\mathrm{p}} = y_0 + h\left(1 + \frac{1}{2}\nabla + \frac{5}{12}\nabla^2 + \frac{3}{8}\nabla^3 + \frac{251}{720}\nabla^4 + \cdots\right) f_0 \qquad (8.24)$$

This is called *Adams–Bashforth* formula and is used as a *predictor* formula (the superscript p indicating that it is a predicted value).

A corrector formula can be derived in a similar manner by using Newton's backward difference formula at f_1:

$$f(x, y) = f_1 + n\nabla f_1 + \frac{n(n+1)}{2}\nabla^2 f_1 + \frac{n(n+1)(n+2)}{6}\nabla^3 f_1 + \cdots \qquad (8.25)$$

Substituting Eq. (8.25) in Eq. (8.23), we obtain

$$y_1 = y_0 + \int_{x_0}^{x_1} \left[f_1 + n\nabla f_1 + \frac{n(n+1)}{2}\nabla^2 f_1 + \cdots \right] dx$$

$$= y_0 + h \int_{1}^{0} \left[f_1 + n\nabla f_1 + \frac{n(n+1)}{2}\nabla^2 f_1 + \cdots \right] dn$$

$$= y_0 + h\left(1 - \frac{1}{2}\nabla - \frac{1}{12}\nabla^2 - \frac{1}{24}\nabla^3 - \frac{19}{720}\nabla^4 - \cdots\right) f_1 \qquad (8.26)$$

The right side of Eq. (8.26) depends on y_1, y_0, y_{-1}, … where for y_1 we use y_1^{p}, the predicted value obtained from (8.24). The new value of y_1 thus obtained from Eq. (8.26) is called the *corrected* value, and hence we rewrite the formula as

$$y_1^{\mathrm{c}} = y_0 + h\left(1 - \frac{1}{2}\nabla - \frac{1}{12}\nabla^2 - \frac{1}{24}\nabla^3 - \frac{19}{720}\nabla^4 - \cdots\right) f_1^{\mathrm{p}} \qquad (8.27)$$

This is called *Adams–Moulton corrector* formula the superscript c indicates that the value obtained is the corrected value and the superscript p on the right indicates that the predicted value of y_1 should be used for computing the value of $f(x_1, y_1)$.

In practice, however, it will be convenient to use formulae (8.24) and (8.27) by ignoring the higher-order differences and expressing the lower-order differences in terms of function values. Thus, by neglecting the fourth and higher-order differences, formulae (8.24) and (8.27) can be written as

$$y_1^{\mathrm{p}} = y_0 + \frac{h}{24}(55 f_0 - 59 f_{-1} + 37 f_{-2} - 9 f_{-3}) \qquad (8.28)$$

and

$$y_1^{\mathrm{c}} = y_0 + \frac{h}{24}(9 f_1^{\mathrm{p}} + 19 f_0 - 5 f_{-1} + f_{-2}) \qquad (8.29)$$

in which the errors are approximately

$$\frac{251}{720} h^5 f_0^{(4)} \quad \text{and} \quad -\frac{19}{720} h^5 f_0^{(4)} \quad \text{respectively.}$$

The general forms of formulae (8.28) and (8.29) are given by

$$y_{n+1}^{\mathrm{p}} = y_n + \frac{h}{24}[55 f_n - 59 f_{n-1} + 37 f_{n-2} - 9 f_{n-3}] \qquad (8.28a)$$

and

$$y_{n+1}^{\mathrm{c}} = y_n + \frac{h}{24}[9 f_{n+1}^{\mathrm{p}} + 19 f_n - 5 f_{n-1} + f_{n-2}] \qquad (8.29a)$$

Such formulae, expressed in ordinate form, are often called *explicit predictor–corrector* formulae.

The values y_{-1}, y_{-2} and y_{-3}, which are required on the right side of Eq. (8.28) are obtained by means of the Taylor's series, or Euler's method, or Runge–Kutta method. Due to this reason, these methods are called *starter* methods. For practical problems, Runge–Kutta fourth-order formula together with formulae (8.28) and (8.29) have been found to be the most successful combination. The following example will illustrate the application of this method.

Example 8.11 We consider once again the differential equation given in Example 8.9 with the same condition, and we wish to compute $y(0.8)$.

For this example, the starter values are $y(0.6)$, $y(0.4)$ and $y(0.2)$, which are already computed in Example 8.9 by the fourth-order Runge–Kutta method. Using now Eq. (8.28) with $y_0 = 0.6841$, $y_{-1} = 0.4228$, $y_{-2} = 0.2027$ and $y_{-3} = 0$, we obtain

$$\begin{aligned} y^{\mathrm{p}}(0.8) &= 0.6841 + \frac{0.2}{24}\{55[1+(0.6841)^2] - 59[1+(0.4228)^2] \\ &\quad + 37[1+(0.2027)^2] - 9\} \\ &= 1.0233, \text{ on simplification.} \end{aligned}$$

Using this predicted value on the right side of Eq. (8.29), we obtain

$$\begin{aligned} y^{\mathrm{c}}(0.8) &= 0.6841 + \frac{0.2}{24}\{9[1+(0.0233)^2] + 19[1+(0.6841)^2] \\ &\quad - 5[1+(0.4228)^2] + [1+(0.2027)^2]\} \\ &= 1.0296, \text{ which is correct to four decimal places} \end{aligned}$$

The importance of the method lies in the fact that when once y_1^{p} is computed from formula (8.28), formula (8.29) can be used iteratively to obtain the value of y_1 to the accuracy required.

8.6.2 Milne's Method

This method uses Newton's forward difference formula in the form

$$f(x, y) = f_0 + n\Delta f_0 + \frac{n(n-1)}{2}\Delta^2 f_0 + \frac{n(n-1)(n-2)}{6}\Delta^3 f_0 + \cdots \qquad (8.30)$$

Substituting Eq. (8.30) in the relation

$$y_4 = y_0 + \int_{x_0}^{x_4} f(x, y)\, dx \qquad (8.31)$$

Hence (i) simplifies to

$$-2\int_0^1 t\,dx + \int_0^1 tx(1-x)dx + \int_0^1 x^2(1-x)dx = 0$$

$$\Rightarrow \qquad -2\int_0^1 \alpha_1 x(1-x)dx + \int_0^1 \alpha_1 x^2(1-x)^2\,dx + \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 = 0$$

$$\Rightarrow \qquad \alpha_1 = \frac{5}{18} = 0.2778, \text{ an simplification.}$$

Then a first approximation to the solution is

$$y(0.5) = \frac{5}{18}(0.5)(0.5) = 0.06944.$$

The exact solution to the given boundary value problem is

$$y(x) = \frac{\sin x}{\sin 1} - x,$$

which means that our solution has an error of 0.0003.

The above approximation can be improved by assuming that

$$t(x) = \alpha_1 x(1 - x) + \alpha_2 x^2(1 - x).$$

Proceeding as above, we obtain

$$\alpha_1 = 0.1924 \text{ and } \alpha_2 = 0.1707.$$

It is clear that by adding more terms to $t(x)$, we can obtain the result to the desired accuracy.

EXERCISES

8.1. Given

$$\frac{dy}{dx} = 1 + xy, \quad y(0) = 1,$$

obtain the Taylor series for $y(x)$ and compute $y(0.1)$, correct to four decimal places.

8.2 Show that the differential equation

$$\frac{d^2y}{dx^2} = -xy, \quad y(0) = 1 \text{ and } y'(0) = 0,$$

has the series solution

$$y = 1 - \frac{x^3}{3!} + \frac{1\times 4}{6!}x^6 - \frac{1\times 4\times 7}{9!}x^9 + \cdots$$

8.3 If

$$\frac{dy}{dx} = \frac{1}{x^2 + y} \quad \text{with } y(4) = 4,$$

compute the values of $y(4.1)$ and $y(4.2)$ by Taylor's series method.

8.4 Use Picard's method to obtain a series solution of the problem given in Problem 8.1 above.

8.5 Use Picard's method to obtain $y(0.1)$ and $y(0.2)$ of the problem defined by

$$\frac{dy}{dx} = x + yx^4, \quad y(0) = 3.$$

8.6 Using Euler's method, solve the following problems:

(a) $\frac{dy}{dx} = \frac{3}{5}x^3 y, y(0) = 1$ (b) $\frac{dy}{dx} = 1 + y^2, y(0) = 0$

8.7 Compute the values of $y(1.1)$ and $y(1.2)$ using Taylor's series method for the solution of the problem

$$y'' + y^2 y^1 = x^3, \ y(1) = 1 \text{ and } y'(1) = 1.$$

8.8 Find, by Taylor's series method, the value of $y(0.1)$ given that

$$y'' - xy' - y = 0, \ y(0) = 1 \text{ and } y'(0) = 0.$$

8.9 Using Picard's method, find $y(0.1)$, given that

$$\frac{dy}{dx} = \frac{y - x}{y + x} \quad \text{and} \quad y(0) = 1.$$

8.10 Using Taylor's series, find $y(0.1)$, $y(0.2)$ and $y(0.3)$ given that

$$\frac{dy}{dx} = xy + y^2, \quad y(0) = 1.$$

8.11 Given the differential equation

$$\frac{dy}{dx} = x^2 + y$$

with $y(0) = 1$, compute $y(0.02)$ using Euler's modified method.

8.12 Solve, by Euler's modified method, the problem

$$\frac{dy}{dx} = x + y, \ y(0) = 0.$$

Choose $h = 0.2$ and compute $y(0.2)$ and $y(0.4)$.

8.13 Given the problem

$$\frac{dy}{dx} = f(x, y) \text{ and } y(x_0) = y_0,$$

an approximate solution at $x = x_0 + h$ is given by the third order Runge–Kutta formula

$$y(x_0 + h) = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3) + R_4,$$

where

$$k_1 = hf(x_0, y_0), \quad k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right)$$

and $$k_3 = hf(x_0 + h, y_0 + 2k_2 - k_1).$$

Show that R_4 is of order h^4.

8.14 Write an algorithm to implement Runge–Kutta fourth order formula for solving an initial value problem.

Find $y(0,1)$, $y(0.2)$ and $y(0.3)$ given that

$$y' = 1 + \frac{2xy}{1+x^2}, \quad y(0) = 0$$

8.15 Use Runge–Kutta fourth order formula to find $y(0.2)$ and $y(0.4)$ given that

$$y' = \frac{y^2 - x^2}{y^2 + x^2}, \quad y(0) = 1.$$

8.16 Solve the initial value problem defined by

$$\frac{dy}{dx} = \frac{3x + y}{x + 2y}, \quad y(1) = 1$$

and find $y(1.2)$ and $y(1.4)$ by the Runge–Kutta fourth order formula.

8.17 State Adam's predictor-corrector formulae for the solution of the equation

$$y' = f(x, y), \; y(x_0) = y_0.$$

Given the problem

$$y' + y = 0, \; y(0) = 1,$$

find $y(0.1)$, $y(0.2)$, and $y(0.3)$ by Runge–Kutta fourth order formula and hence obtain $y(0.4)$ by Adam's formulae.

8.18 Given the initial value problem defined by

$$\frac{dy}{dx} = y(1 + x^2), \quad y(0) = 1$$

find the values of y for $x = 0.2$, 0.4, 0.6, 0.8 and 1.0 using the Euler, the modified Euler and the fourth order Runge–Kutta methods. Compare the computed values with the exact values.

8.19 State Milne's predictor-corrector formulae for the solution of the problem

$$y' = f(x, y), \; y(x_0) = y_0.$$

Given the initial value problem defined by

$$y' = y^2 + xy, \; y(0) = 1,$$

find, by Taylor's series, the values of $y(0.1)$, $y(0.2)$ and $y(0.3)$. Use these values to compute $y(0.4)$ by Milne's formulae.

8.20 Using Milne's formulae, find $y(0.8)$ given that

$$\frac{dy}{dx} = x - y^2, \quad y(0) = 0, \quad y(0.2) = 0.02,$$

$y(0.4) = 0.0795$ and $y(0.6) = 0.1762$.

8.21 Explain what is meant by a fourth order formula. Discuss this with reference to the solution of the problem

$$\frac{dy}{dx} = 3x + \frac{y}{2}, \quad y(0) = 1$$

by Runge–Kutta fourth order formula.

8.22 Use Taylor's series method to solve the system of differential equations

$$\frac{dx}{dt} = y - t, \quad \frac{dy}{dt} = x + t$$

with $x = 1$, $y = 1$ when $t = 0$, taking $\Delta x = \Delta t = 0.1$.

8.23 Using fourth order Runge–Kutta method, compute the value of $y(0.2)$ given that

$$\frac{d^2y}{dx^2} + y = 0$$

with $y(0) = 1$ and $y'(0) = 0$.

8.24 Given that

$$y'' - xy' + 4y = 0, \quad y(0) = 3, \; y'(0) = 0,$$

compute the value of $y(0.2)$ using Runge–Kutta fourth order formla.

8.25 Solve the boundary value problem defined by

$$y'' - y = 0, \quad y(0) = 0, \quad y(1) = 1,$$

by finite difference and cubic spline methods. Compare the solutions obtained at $y(0.5)$ with the exact value. In each case, take $h = 0.5$ and $h = 0.25$.

8.26 *Shooting method* This is a popular method for the solution of two-point boundary value problems. If the problem is defined by

$$y'' = f(x), \quad y(x_0) = 0 \text{ and } y(x_1) = A,$$

then it is first transformed into the initial value problem

$$y'(x) = z, \quad z'(x) = f(x),$$

with $y(x_0) = 0$ and $z(x_0) = m_0$, where m_0 is a guess for the value of $y'(x_0)$. Let the solution corresponding to $x = x_1$ be Y_0. If Y_1 is the value obtained by another guess m_1 for $y'(x_0)$, then Y_0 and Y_1 are related linearly. Thus, linear interpolation can be carried out between the values (m_0, y_0) and (m_1, y_1).

Obviously, the process can be repeated till we obtain a value for $y(x_1)$ which is close to A.

Apply the shooting method to solve the boundary value problem

$$y'' = y(x),\ y(0) = 0 \text{ and } y(1) = 1.$$

8.27 Fyfe [1969] discussed the solution of the boundary value problem defined by

$$y'' + \frac{4x}{1+x^2}y' + \frac{2}{1+x^2}y = 0, \quad y(0) = 1 \text{ and } y(2) = 0.2.$$

Solve this problem by cubic spline method first with $h = 1$ and then with $h = 1/2$ to determine the value of $y(1)$. Compare your results with the exact values of $y(1)$ obtained from the analytical solution $y = 1/(1 + x^2)$.

8.28 *Method of Linear Interpolation* Let the boundary value problem be defined by

$$y'' + f(x)y' + g(x)y = p(x),$$
$$y(x_0) = y_0 \quad \text{and} \quad y(x_n) = y_n.$$

Set up the finite difference approximation of the differential equation and solve the algebraic equations using the initial value y_0 and assuming a value, say Y_0, for $y(x_1)$. Again, we assume another value for $y(x_1)$, say Y_1 and then compute the values of $y_2, y_3, \ldots, y_{n-1}$ and y_n. We, thus, have two sets of values of $y(x_1)$ and $y(x_n)$. Now we use linear interpolation formula to compute the value of $y(x_1)$ for which $y(x_n) = y_n$. The process is repeated until we obtain the value of $y(x_n)$ close to the given boundary condition (see Problem 8.26).

Solve the boundary value problem defined by

$$y'' + xy' - 2y = 0,\ y(0) = 1 \text{ and } y(1) = 2$$

using the method of Linear interpolation.

8.29 Using Galerkin's method, compute the value of $y(0.5)$ given that

$$y'' + y = x^2,\ 0 < x < 1, \quad y(0) = 0 \text{ and } y(1) = 0.$$

8.30 Solve Poisson's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2, \quad 0 \le x, y \le 1$$

with $u = 0$ on the boundary C of the square region $0 \le x \le 1$, $0 \le y \le 1$.

Answers to Exercises

8.1 1.1053

8.2 $1 - \frac{x^3}{3!} + \frac{1\times 4}{6!}x^6 - \frac{1\times 4\times 7}{9!}x^9 + \cdots$

8.3 4.005, 4.0098

8.4 $1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \cdots$

8.5 3.005, 3.0202

8.6 (a) 1.0006, (b) $y_1 = 1.0000$, $y_2 = 0.201$, $y_3 = 0.3020$

8.7 1.1002, 1.2015

8.8 1.005012

8.9 1.0906

8.10 1.11686, 1.27730, 1.5023

8.11 1.0202

8.12 0.0222, 0.0938

8.14 0.1006, 0.2052, 0.3176

8.15 0.19598, 1.3751

8.16 1.2636, 1.532

8.17 $y(0.1) = 0.90484$, $y(0.2) = 0.81873$, $y(0.3) = 0.7408$
$y(0.4) = 0.6806$ (Exact value = 0.6703).

8.18 $x = 0.2$ 0.024664 (Euler)
0.003014 (Modified Euler)
0.000003 (Runge–Kutta)
$x = 1.0$ 0.776885 (Euler)
0.12157 (Modified Euler)
0.000273 (Runge–Kutta)

8.19 $y(0.1) = 1.1169$, $y(0.2) = 1.2773$, $y(0.3) = 1.5023$,
$y(0.4) = 1.8376$.

8.20 $y^p(0.8) = 0.3049$, $y^c(0.8) = 0.30460$

8.21 $h = 0.2$, error = 0.00000110
$h = 0.1$, error = 0.0000007

8.22 $x_1 = 1.1003$, $y(0.1) = y_1 = 1.1100$

8.23 0.980067 (Exact value = 0.980066)

8.24 $y(0.2) = 2.762239$ (Exact value = 2.7616)
$z(0.2) = -2.360566$ (Exact value = –2.368)

8.25 Exact value of $y(0.5) = 0.443409$
(a) 0.443674, (b) 0.443140

8.26 $y'(0) = 2.8$

8.27 (a) $h = 1, \quad y_1 = 0.4$

$$h = \frac{1}{2}, \quad y_2 = 0.485714$$

(b) $h = 1, \quad y_1 = 0.542373$

$$h = \frac{1}{2}, \quad y_2 = 0.5228$$

8.28 $y_1 = 1.0828, \quad y_2 = 1.2918, \quad y_3 = 1.6282, \quad y_4 = 1.99997$

8.29 $y(0.5) = -0.041665$ (Exact value = –0.04592)

8.30 $u(x, y) = -\frac{5}{2} xy(x - 1)(y - 1)$